

LECTURE 1.1



Data and Variation

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

What a Year of Schooling Is Worth

Put a number on it. Then say whose earnings you would need to see:

Question

How much more does a person earn per hour, on average, for each extra year they stayed in school?

Lecture Plan

1 How This Module Works.

What the term covers, how it is assessed, and the standing question.

2 Populations and Samples.

The question is about a population; the data is a sample.

3 Describing One Variable.

Centre, spread and shape, and what each one hides.

4 Describing Two Variables.

Covariance, correlation, and what one number cannot show.

SECTION 1

How This Module Works

1. How This Module Works
2. Populations and Samples
3. Describing One Variable
4. Describing Two Variables

What the Module Covers: The Linear Regression Model

The module does one thing thoroughly: the linear regression model. The term runs from what a sample is to reading a residual plot.

Run One, and Name Its Assumption

You can run a regression and say what its coefficient hides. You can name the assumption that would license a causal reading.

How the Four Blocks Fit Together

- **Foundations:** what a sample is, and why one estimate is only one draw.
- **Simple Regression:** one line, its derivation, and how far to trust it.
- **Multiple Regression:** what holding other things fixed buys, and what omitting them costs.
- **Limits:** what breaks, how you would know, and what regression cannot license.

Simple Before Multiple Regression

Lecture 3.1 is unreadable without 2.1 to 2.4. Block 4 is where the module stops being mechanical.

The Question We Ask Every Week

Regression says how two things vary together. Whether co-movement licenses a causal claim depends on the design, not the estimator.

The Standing Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Nothing Is Derived in This Room

- **The Result:** what it involves and what it buys you, in the lecture.
- **The Derivation:** in the appendix of its deck, and set as homework.
- **The Solutions:** published with the problem set, after you have tried it.

Watching a Proof Is Not Learning It

A derivation copied off a screen teaches nothing you can reproduce. You get the shape of it here and fill in the steps yourself.

Assessment: Components, Weights and Weekly Workload

[BLOCKED – needs the module descriptor. One slide: component, weight, week, what it tests, which outcome it evidences, and one AI-policy line each.]

Nothing Here Is Read Aloud

Everything not on this frame is in the outline and the appendix. A room that hears a document read aloud concludes the module is low-stakes.

SECTION 2

Populations and Samples

1. How This Module Works
2. Populations and Samples
3. Describing One Variable
4. Describing Two Variables

The Gap Between the Population and the Sample

Two groups of people, and the whole module is the argument between them:

- **Population:** every person the question is about.
- **Sample:** the handful of those people whose numbers we hold.
- **The Whole Module:** an argument from the sample back to the population.

No Technique Closes the Gap

Nothing later in the term removes the gap. Later techniques only say honestly how wide the gap might be.

Notation for a Single Observation

A sample is a list of pairs, one for each person in it:

$$\{(x_i, y_i)\}_{i=1}^n \quad (1)$$

- x_i : years of schooling completed by person i
- y_i : hourly earnings of person i , in euro
- i : which person, running from 1 to n
- n : how many people are in the sample

Cross-Section, Time Series and Panel Data

What you can estimate depends on how the data is indexed:

- **Cross-Section Data:** many people, each seen once, indexed by i .
- **Time-Series Data:** one unit, seen over many periods, indexed by t .
- **Panel Data:** many people, each over many periods, indexed by i and t .

Every Dataset Here Is Cross-Section

Our 120 people are each observed once. The assumptions change when the subscript does.

Roman for What We Know, Greek for What We Want to Know

Roman here means the everyday alphabet, the one English is written in. Our sample holds n people; the population holds N .

Never the Same Number

Every summary from here comes as a pair: a Roman letter computed from our 120, and a Greek letter for the fact about all 200,000 that nobody sees.

The Population We Will Use All Term

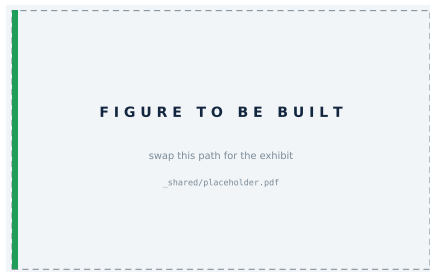


Figure 1: Schooling and hourly earnings: the population in grey, the two samples in colour.

A Year Pays 1.30 Here

We set the generator so a year of schooling pays 1.30 an hour. Twelve years of schooling pays 9.50. Neither number came from data.

Why a Simulated Population Rather Than Real Data

Real earnings data gives an estimate and never says whether it was good. Here the truth is written down. Every estimate this term can be marked against 1.30.

The Flaws Are Already Switched Off

Lecture 3.1 turns on an omitted variable and Lecture 4.2 a mismeasured one. Lecture 4.1 turns on an error whose size grows with schooling.

The Training Sample and the Hold-Out Sample

The 120 we work with is the **training** sample. A second 120 was drawn at the same time and put away unopened. The second draw is the **hold-out**, or test, sample. Statistics and machine learning both call the pair a train-test split (James et al., 2021).

A Fit Proves Nothing Alone

Anything can be made to fit the data it was chosen on. The only honest check is data the choice never saw.

SECTION 3

Describing One Variable

1. How This Module Works
2. Populations and Samples
3. Describing One Variable
4. Describing Two Variables

The Mean Is Where the Distribution Balances

The mean is the one number that stands in for everybody. It is the point where deviations above and below sum to zero:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (2)$$

- $\sum_{i=1}^n$: add up what follows, over all n people in the sample
- $\bar{x}$: said “x-bar”, where the bar means the average of x
- μ_x : the average of x across the population

Why the Bar Marks a Sample Quantity

A bar over a letter means the average of that letter. Statistics marks sample averages this way everywhere, not only here. Our 120 earn 12.25 an hour on average; the population earns 11.52.

Redraw and the Bar Changes

$\bar{x}$ is arithmetic you can do this afternoon on 120 numbers. μ_x is a fact about 200,000 people that you will never compute. Draw a different 120 and $\bar{x}$ changes; μ_x does not.

Expectation: The Population Average of an Expression

μ_x is the population average of x itself. We also want the population average of things built from x :

An Operator, Not a Quantity

$E[\cdot]$ takes whatever expression sits inside it and returns its population average. $E[x]$ is μ_x ; feed it something else and you get something else.

Lecture 1.2 is about this operator. Today it is only notation.

Mean, Median and Mode

Three answers to where a distribution sits. They agree only when it is symmetric:

- **Mean:** the balance point, and the one a long tail pulls.
- **Median:** the middle value, with half the people either side.
- **Mode:** the most common value, at the peak.

The letters tell median and mode apart. The i is for middle, the o for most common.

Earnings Push the Mean Above the Median

A town's earnings: the median is 26,000 and the mean 34,000. The peak sits lower still. Report the mean and you describe a person who may not exist.

The Three Centres on Two Distributions

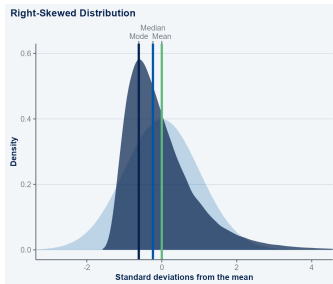


Figure 2: A long right tail: the three separate.

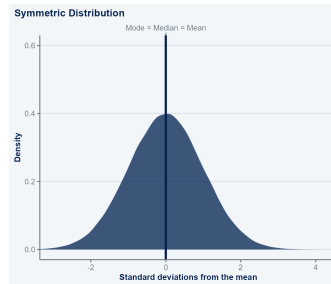


Figure 3: No tail either way: the three coincide.

A Symmetric Distribution Has One Centre

Navy marks the mode, blue the median, green the mean. On a symmetric distribution all three sit at the same point.

The Variance: Average Squared Distance From the Mean

A mean tells you nothing about whether people cluster near it or scatter. The variance measures how far a typical person sits from the mean:

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (3)$$

- s_x^2 : the variance of x in the sample
- σ_x^2 : the variance of x in the population

Bessel's Correction

An average weights the sum by $1/n$. Bessel's correction weights it by $1/(n - 1)$ instead. Deviations from the sample mean sum to zero, so only $n - 1$ are free. The weight moves sharply in small samples:

- $n = 120$: $1/119$ against $1/120$, a difference under one per cent.
- $n = 2$: the weight is 1, so the sum of squares is the estimate.
- $n = 1$: the weight is $1/0$, so there is no estimate at all.

The Standard Deviation

The variance is in squared euro, and nothing is measured in those. Taking the positive square root puts the spread back in euro:

$$s_x = \sqrt{s_x^2} \quad (4)$$

- s_x : the standard deviation of x in the sample
- σ_x : the standard deviation of x in the population

Read It Against the Mean

Our 120 average 12.25 an hour, with a standard deviation of 5.91. Double everybody's earnings and both numbers double; the variance quadruples.

Comparing Centre and Spread Across Distributions

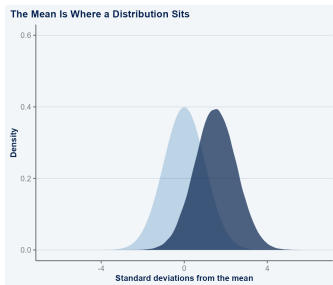


Figure 4: The mean: same spread, different centre.

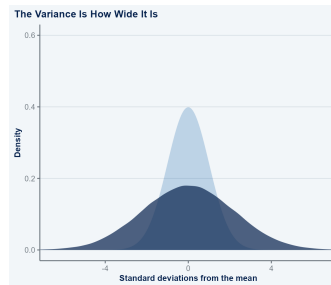


Figure 5: The variance: same centre, different spread.

Centre and Spread Move Independently

Grey is the same reference curve in both panels. The mean moves in one panel and the spread in the other. Neither touches the shape.

The Moments of a Distribution

Mean, variance, skewness and kurtosis are the first four moments of a distribution. The mean locates it; the rest are one bracket at three powers:

$$E \left[(x - \mu_x)^k \right], \quad k = 2, 3, 4 \quad (5)$$

- k : the power, giving the variance, then skewness, then kurtosis

A Higher Power Weights the Tails More

Double a person's distance from the mean. At $k = 2$ they count four times more; at $k = 4$, sixteen. Kurtosis is decided by its outliers. An odd power keeps the sign, so $k = 3$ says which tail is longer.

Skewness

Skewness asks which side of the mean has the longer tail. Cube the deviations, then divide by σ_x^3 to strip the units:

$$\text{skew}(x) = \frac{E \left[(x - \mu_x)^3 \right]}{\sigma_x^3} \quad (6)$$

- **Below 0.5:** approximately symmetric.
- **0.5 to 1:** moderately skewed.
- **Above 1:** highly skewed.

Bulmer's Bands Read the Size, Not the Sign

Positive means a long right tail, negative a long left one (Bulmer, 1979). The bands apply to the number with the sign stripped off.

Skewness: Which Tail Runs Longer

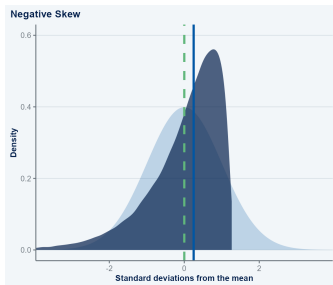


Figure 6: Negative skew: the long tail runs left.

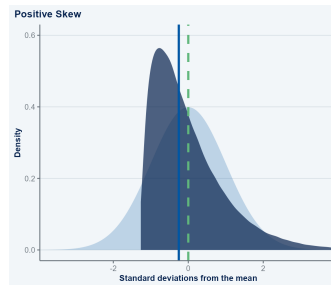


Figure 7: Positive skew: the long tail runs right.

The Tail Pulls the Mean

This is the third power: both curves share the mean and the variance. Grey marks the median, green the mean, which sits out in the tail.

Kurtosis

Kurtosis asks how much of the distribution sits far from the mean. Raise the deviations to the fourth power, then divide by σ_x^4 :

$$\text{kurt}(x) = \frac{E \left[(x - \mu_x)^4 \right]}{\sigma_x^4} \quad (7)$$

- **Below 3:** thinner tails than the normal curve.
- **Exactly 3:** the normal curve itself.
- **Above 3:** thicker tails, and more weight in them.

No Agreed Bands, Only the Benchmark

Nobody has settled bands for kurtosis the way Bulmer did for skewness. Daily stock returns sit well above three, which is why finance cares about it.

Kurtosis: How Much Sits Far From the Middle

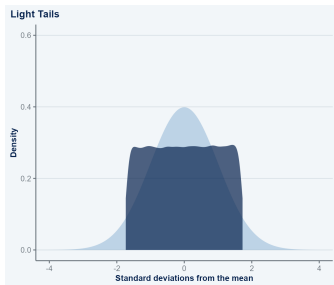


Figure 8: Light tails: almost nothing lands far out.

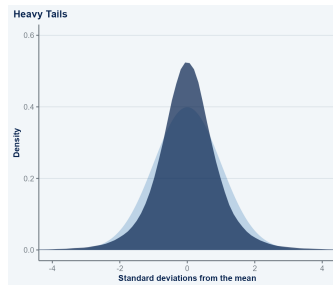


Figure 9: Heavy tails: much more lands a long way out.

Thicker Tails, Same Variance

This is the fourth power: both panels share the normal curve's mean and variance. What changes is how often a value turns up far out.

The Range

The range is the measure everybody reaches for first. It is also the only one here that is not a moment:

$$\text{range} = x_{\max} - x_{\min} \quad (8)$$

Two People Decide the Range

Every other observation could move without changing it. A bigger sample can only widen the range, never narrow it.

SECTION 4

Describing Two Variables

1. How This Module Works
2. Populations and Samples
3. Describing One Variable
4. Describing Two Variables

Covariance: Measuring How Two Variables Move Together

Each variable has its own mean. Every person has a distance from each, and the covariance multiplies the two:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y}) \quad (9)$$

- s_{xy} : the covariance of x and y in the sample

Same Side Positive, Opposite Sides Negative

Above on both, or below on both, and the product is positive. Above on one and below on the other, and it is negative.

The Sign Is the Part You Can Read

Sum those products across all 120 people, and ours comes out positive:

- **Positive:** above-average schooling mostly comes with above-average earnings.
- **Negative:** above-average schooling mostly comes with below-average earnings.
- **Near Zero:** the products cancel, so no straight-line relationship shows.

Zero Is Cancellation, Not a Convention

Zero does not mean unrelated by definition. It means the positive products and the negative ones came out even.

Why the Size of a Covariance Is Not Interpretable

The same 120 people and the same relationship, measured three ways:

- **Earnings in Euro an Hour:** 8.11 euro-years
- **Earnings in Cents an Hour:** 811.16 cent-years
- **Schooling in Months Instead:** 97.34 euro-months

Only the Yardstick Changed

Nothing about these 120 people changed from one line to the next. Only the units did, so the size of a covariance carries no information.

Correlation Is Covariance With the Units Divided Out

The covariance carries euro-years; so does the product of the two standard deviations. Divide and the units cancel:

$$r_{xy} = \frac{s_{xy}}{s_x s_y} \quad (10)$$

- r_{xy} : the correlation of x and y in the sample

All Three Become 0.594

Every one of those three covariances divides down to 0.594. Both standard deviations are positive, so the covariance's sign survives.

Correlation Runs From Minus One to One

Correlation says how strong the relationship is, not only which way it runs. Price and quantity demanded move opposite ways, so that correlation is negative.

One Means an Exact Line

At 1 or -1 the points lie exactly on a straight line. Values nearer either end mean a stronger straight-line relationship. Measure earnings in cents and nothing moves.

Small, Medium and Large: 0.1, 0.3, 0.5

Cohen's conventions, the ones you will meet everywhere (Cohen, 1988):

- $r = 0.1$: small.
- $r = 0.3$: medium.
- $r = 0.5$: large.

Cohen Wrote These for Psychology

Wooldridge gives no bands at all. Ours is 0.594, large by Cohen. Squared, it says the two share a third of their variation.

Correlations of 0.10, 0.30 and 0.50

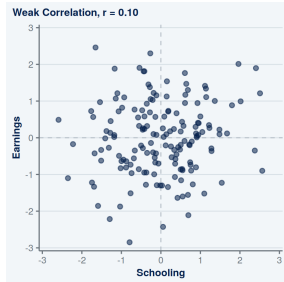


Figure 10: Cohen's small.

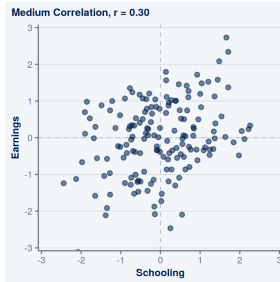


Figure 11: Cohen's medium.

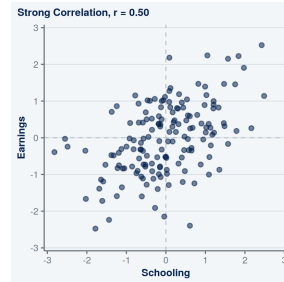


Figure 12: Cohen's large.

Earnings Depend on More Than Schooling

Even the panel Cohen calls large is a wide cloud. Earnings depend on far more than schooling, which is what block 3 adds.

Correlation Measures Straight-Line Association Only

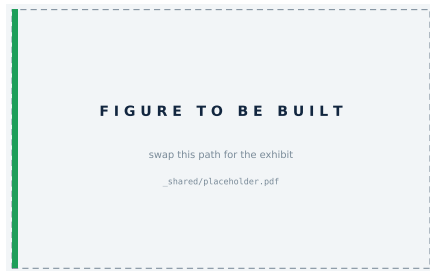


Figure 13: Four samples from this population, each with a correlation of 0.60.

Same Number, Four Clouds

A curve, four outliers and two separate groups all return 0.60. One statistic cannot tell them apart (Anscombe, 1973; Matejka & Fitzmaurice, 2017).

Why a Correlation Is Not a Causal Claim

Schooling really does raise earnings here by 1.30 an hour a year. We wrote the generator that way. Outside a simulation you are handed the correlation and not the generator.

0.594 Is Consistent With No Effect

The same 0.594 could come from a world where schooling changes nothing. The number alone cannot tell you which world you are in.

Three Ways a Correlation Misleads

All three are real correlations, and each one goes wrong differently:

- **Coincidence:** films starring Nicolas Cage track pool drownings, $r = 0.66$.
- **A Common Cause:** on sunny days people buy ice cream, and people swim.
- **A Difference That Pays:** people who stay in school longer differ in more than schooling.

Ice Cream Does Not Cause Drowning

Nothing connects Cage to drownings at all. Sunshine drives both the ice cream and the swimming. Ability raises schooling and earnings alike. The correlation looks identical in all three.

What Would Have To Be True Today?

Today's number is the correlation between schooling and earnings in our sample.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Application: The Four Numbers on Our Sample

Open the sample of 120 and plot the cloud. Compute the mean, median, standard deviation, and the correlation of schooling with earnings.

Question

We built this population so a year of schooling is worth 1.30. Does that make the correlation close to one?

Three Ways to Describe Data

Today was three separate ideas, each answering a different question:

1 Populations and Samples.

The question is about a population; the gap to the sample never closes.

2 Describing One Variable.

Centre, spread and shape each answer one question, and each throws something away.

3 Describing Two Variables.

Covariance and correlation measure straight-line co-movement, and nothing else.

Before Lecture 1.2

1 Play:

load the sample of 120 in R and compute today's four numbers.

2 Apply:

problem set 1, which opens with the appendix derivation of Bessel's correction.

3 Review:

Wooldridge (2020), Math Refresher B, sections 3 and 4.

Diagnostic in This Week's Lab

Ten anonymous, ungraded items on expectation, sampling, logs and proportions.

Next Week

Instead of one average for everybody, an average for each level of schooling.

An Average at Each Year

The line a regression draws joins one average per year of schooling.

Thank you

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BACKUP

Appendix

Derivation: Bessel's Correction

No other number makes the squared deviations smaller than $\bar{x}$ does. So the expected sum of squared deviations from $\bar{x}$ comes to $(n - 1)\sigma_x^2$, not $n\sigma_x^2$.

The Cross Term Merges Into the Gap

Write each deviation from $\bar{x}$ as a deviation from μ_x minus the gap. Square and sum, and the cross term merges into the squared gap. Take expectations of what is left.

[Full derivation. Set as homework; worked solution published with the set.]

The Greek Alphabet: Alpha to Mu

Say them out loud. A symbol you cannot pronounce is one you will never ask about:

Lower	Upper	Name	Said as	Used here for
α	A	alpha	AL-fa	
β	B	beta	BEE-ta	the regression coefficients, from 2.1
γ	Γ	gamma	GAM-a	
δ	Δ	delta	DEL-ta	
ϵ	E	epsilon	EP-si-lon	a shift; Δ for a change in
ζ	Z	zeta	ZEE-ta	
η	H	eta	EE-ta	
θ	Θ	theta	THEE-ta	
ι	I	iota	eye-OH-ta	
κ	K	kappa	KAP-a	
λ	Λ	lambda	LAM-da	
μ	M	mu	MEW	a population mean

The Greek Alphabet: Nu to Omega

Lower	Upper	Name	Said as	Used here for
ν	N	nu	NEW	
ξ	Ξ	xi	KSY	
o	O	omicron	OH-mi-kron	
π	Π	pi	PY	
ρ	P	rho	ROH	
σ	Σ	sigma	SIG-ma	a population spread; Σ means add up
τ	T	tau	TAW	
v	Υ	upsilon	UP-si-lon	
ϕ	Φ	phi	FY	
χ	X	chi	KY	
ψ	Ψ	psi	SY	
ω	Ω	omega	OH-mig-a	

Thirteen capitals are identical to ours, so nobody uses them as symbols. A capital alpha is just an A.

How to Study This Subject

- **Re-derive:** the week's central result, from a blank page, before the tutorial.
- **Attempt:** the problem set before reading the answers, not after.
- **Space:** the work across the week, since four short days beat one long.
- **Avoid:** rereading and highlighting, which give fluency without retrieval.

Nothing on This Frame Is Examinable

Ten minutes today, and no time after that. The four habits above are worth more than any revision guide.

Notes and Tools

- **Wooldridge:** chapter 1 for the subject, Math Refreshers B and C for today (Wooldridge, 2020).
- **Angrist and Pischke:** section 6.1, on why the schooling question fills a term (Angrist & Pischke, 2014).
- **The Train-Test Split:** chapter 2 of James et al. (2021), stated properly.
- **Pictures First:** Healy (2018) on the plot before the statistic, Wickham et al. (2023) for the R.
- **Software:** R and RStudio, both free. [Install links and the module page.]

References

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